

APPLIED OPERATING SYSTEM



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MODULE 1

INTRODUCTION TO OPERATING SYSTEM



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OBJECTIVES

Upon completion of this module, the student will be able to:

- Define the operating system (OS) and its goals;
- Demonstrate the abstract view of the components of a computer system;
- Enumerate the types of OS and services provided by the OS;
- Explain the organization and operation of computer system;
- Discuss interrupt, its types and interrupt handling;
- Give different computer architectures, processor systems and computing environments;

OPERATING SYSTEM OVERVIEW



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WHAT IS AN OPERATING SYSTEM?

- An **Operating System (OS)** is a program or system software that acts as an interface between the user and the computer hardware and controls the execution of all kinds of programs.
- It is not possible for the user to use any computer or mobile device without having an operating system.



GOALS OF OPERATING SYSTEM

1. Execute user programs and make solving user problems easier.
2. Make the computer system convenient to use.
3. Use the computer hardware in an efficient manner.



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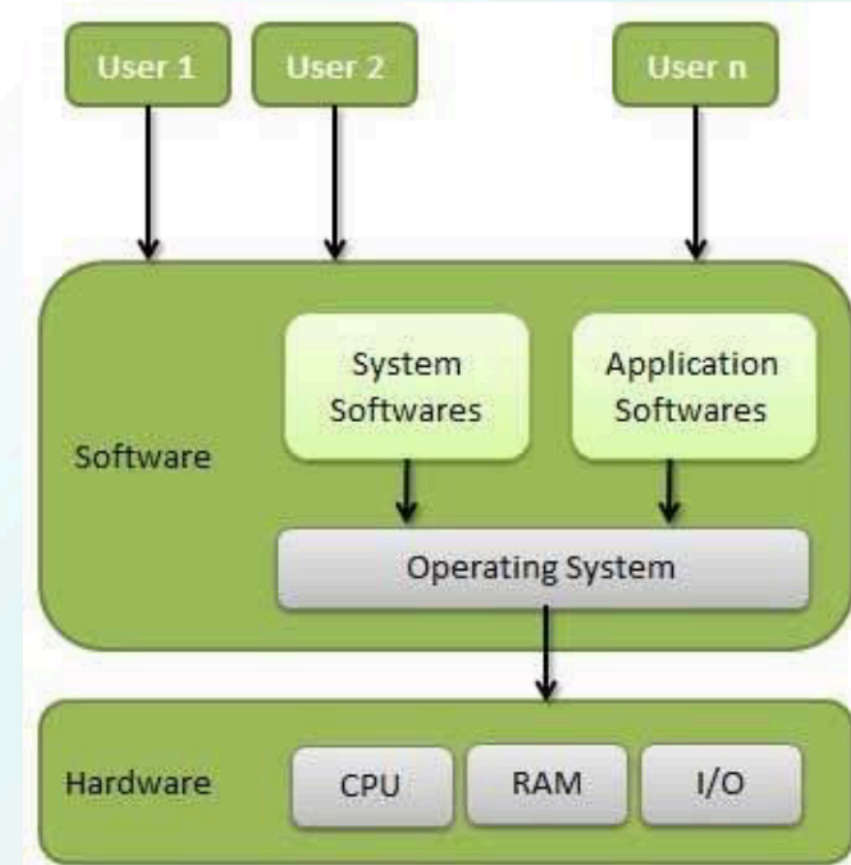


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COMPONENTS OF A COMPUTER SYSTEM

- **Computer hardware** – CPU, memory and I/O devices, provides the basic computing resources.
- **Application programs** – are used to solve the computing problems of the users such as word processors, games and business programs .
- **Users** – who utilize a computer or network service trying to solve different problems.
- **Operating Systems** – controls and coordinates the use of the hardware among the various application programs for the various users.



COMMON SERVICES OFFERED BY ALMOST ALL OPERATING SYSTEMS:

1. User Interface
2. Program Execution
3. File system manipulation
4. Input / Output Operations
5. Communication
6. Resource Allocation
7. Error Detection
8. Accounting
9. Security and protection



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COMMON SERVICES OFFERED BY ALMOST ALL OPERATING SYSTEMS

1. User Interface (UI).

Refers to the part of an OS, or device that allows a user to enter and receive information.

Types of UI:

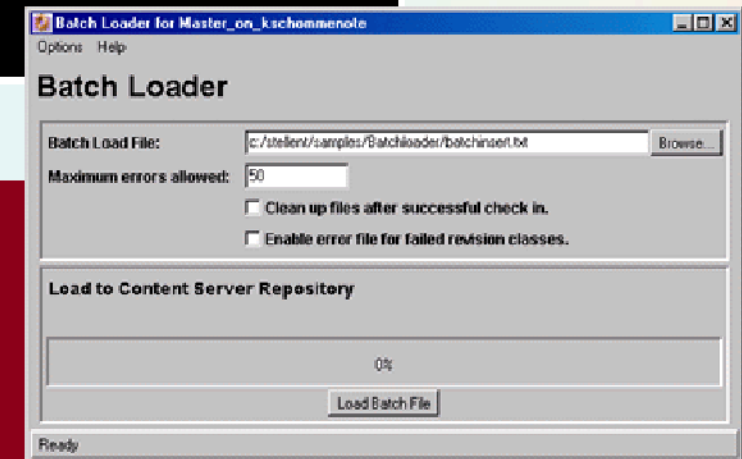
- Command line interface
- Batch based interface
- Graphical User Interface

```
C:\>dir
Volume in drive C is Windows
Volume Serial Number is B4F7-C120

Directory of C:\

01/10/2007  19:38    <DIR>          25c7843ff7527e04ea3749
19/09/2007  16:56    <DIR>          97a61370812ce95093c63d
19/09/2007  16:42                0 AUTOEXEC.BAT
19/09/2007  16:36                211 boot.ini.comodofirewa
19/09/2007  16:42                0 CONFIG.SYS
19/09/2007  16:48    <DIR>          Documents and Settings
19/09/2007  16:53    <DIR>          e4a32567d68c0d02f07b4c
19/09/2007  16:54    <DIR>          Intel
01/03/2008  21:48    <DIR>          Program Files
19/09/2007  16:57                516 RHDSetup.log
25/11/2007  13:03    <DIR>          UBCH4Win
13/03/2008  11:32    <DIR>          WINDOWS
                        4 File(s)      727 bytes
                        8 Dir(s)  1,479,877,888 bytes free

C:\>
```



COMMON SERVICES OFFERED BY ALMOST ALL OPERATING SYSTEMS:

- 2. Program Execution.** The OS must have the capability to load a program into memory and execute that program.
- 3. File system manipulation.** Programs need has to be read and then write them as files and directories. File handling portion of OS also allows users to create and delete files by specific name along with extension, search for a given file and / or list file information.



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COMMON SERVICES OFFERED BY ALMOST ALL OPERATING SYSTEMS:

- 4. Input / Output Operations.** A program which is currently executing may require I/O, which may involve file or other I/O device. The OS is responsible for reading and/or writing data from I/O devices such as disks, tapes, printers, keyboards, etc.
- 5. Communication.** Process needs to swap over information with other process. Processes executing on same computer system or on different computer systems can communicate using operating system support.



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COMMON SERVICES OFFERED BY ALMOST ALL OPERATING SYSTEMS:

- 6. **Resource Allocation** The OS manages the different computer resources such as CPU time, memory space, file storage space, I/O devices, etc. and allocates them to different application programs and users.
- 7. **Error Detection** The operating system should be able to detect errors within the computer system (CPU, memory, I/O, or user program) and take the appropriate action.



COMMON SERVICES OFFERED BY ALMOST ALL OPERATING SYSTEMS:

8. **Job Accounting.** OS keeps track of time and resources used by various tasks and users, this information can be used to track resource usage for a particular user or group of user.
9. **Security and Protection.** *Protection* is any mechanism for controlling access of processes or users to resources defined by the OS. *Security* is a defense of the system against internal and external attacks (denial-of-service, worms, viruses, identity theft, theft of service)



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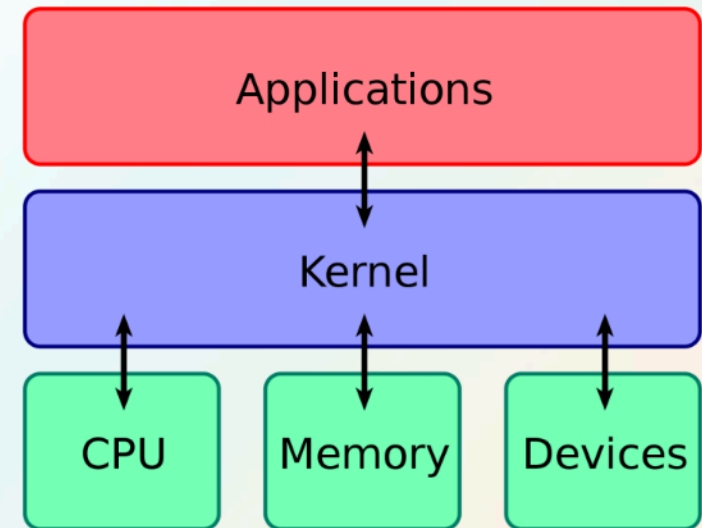
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WHAT IS A KERNEL?

Kernel is the central part of an OS which manages system resources and is always resident in memory. It also acts like a bridge between application and hardware of the computer. It is also the first program that loads after the **bootloader**.

Bootloader is a program that loads and starts the boot time tasks and processes of an OS. It also places the OS of a computer into memory.



TYPES OF OPERATING SYSTEM

Operating systems are there from the very first computer generation and they keep evolving with time.

Types of operating systems which are most commonly used:

1. Batch operating system
2. Time-sharing operating systems
3. Distributed operating system
4. Network operating system
5. Real-time operating system
6. Handheld operating system



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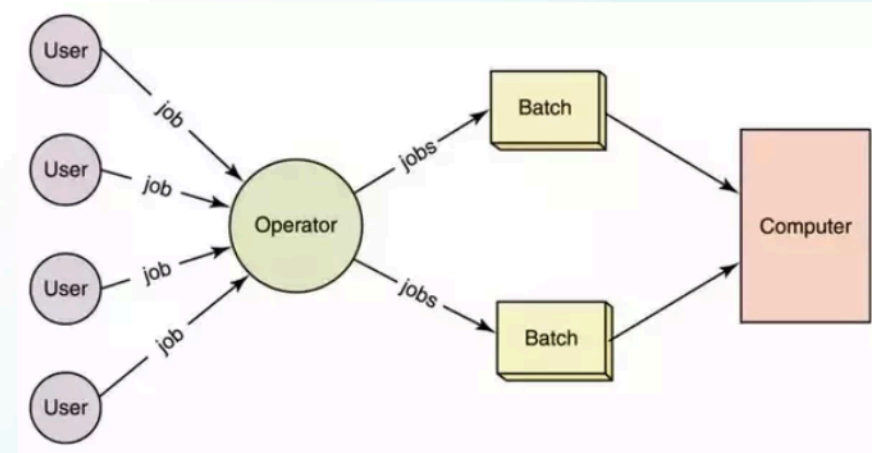
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TYPES OF OPERATING SYSTEM

1. Batch Operating System (BOS)

- The user of a BOS never directly interacts with the computer.
- Every user prepares his or her job on an offline device like a punch card and submit it to the computer operator.
- To speed up processing, jobs with similar needs are batched together and run as a group.
- The programmers leave their programs with the operator then operator sorts the programs with similar requirements into batches.



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TYPES OF OPERATING SYSTEM

2. Time-sharing Operating System

- Time-sharing or **multitasking** is logical extension in which CPU switches jobs so frequently that users can interact with each job while it is running, creating **interactive computing**.
- Multiple jobs are executed by the CPU by switching between them, but the switches occur so frequently. Thus, the user can receive an immediate response. **Response time** should be < 1 second
- Each user has at least one program executing in memory
- If several jobs ready to run at the same time $\Rightarrow$ **CPU scheduling**
- If processes don't fit in memory, **swapping** will take place
- **Examples:** Unix, Linux, Multics and Windows



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TYPES OF OPERATING SYSTEM

3. Distributed Operating System

- Distributed systems use multiple central processors to serve multiple real-time applications and multiple users. Data processing jobs are distributed among the processors accordingly.
- The processors communicate with one another through various communication lines (such as high-speed buses or telephone lines). These are referred as **loosely coupled systems** or **distributed systems**. Processors in a distributed system may vary in size and function. These processors are referred as sites, nodes, computers, and so on.
- **Examples:** Telecom Network, WWW, Cloud Computing, etc.



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TYPES OF OPERATING SYSTEM

4. Network Operating System (NOS)

- A NOS runs on a server and provides the server the capability to manage data, users, groups, security, applications, and other networking functions.
- The primary purpose of the network operating system is to allow shared file and printer access among multiple computers in a network, typically a local area network (LAN), a private network or to other networks.
- **Examples:** Microsoft Windows Server 2003, Microsoft Windows Server 2008, UNIX, Linux, Mac OS X Server, Novell NetWare, and BSD/OS (Berkeley Software Design).



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TYPES OF OPERATING SYSTEM

5. Real-time Operating System (RTOS)

- **RTOS** is an operating system intended to serve **real-time systems/applications** that process data as it comes in, mostly without buffer delay.
- The time interval required to process and respond to inputs is **very small**. This time interval is called **response time**.
- **Real-time systems** are used when there are time requirements are very strict like missile systems, air traffic control systems, robots, etc.
- **Examples:** LynxOS, OSE, QNX, RTLinux, VxWorks, Windows CE



TYPES OF OPERATING SYSTEM

6. Handheld Operating System

- It is also known as **Mobile OS** which is built exclusively for a mobile device, such as a smartphone, personal digital assistant (PDA), tablet, wearable devices or other embedded mobile OS.
- **Examples:** Android, Symbian, iOS, BlackBerry OS and Windows Mobile.



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COMPUTER SYSTEM ORGANIZATION, ARCHITECTURES AND COMPUTING ENVIRONMENT



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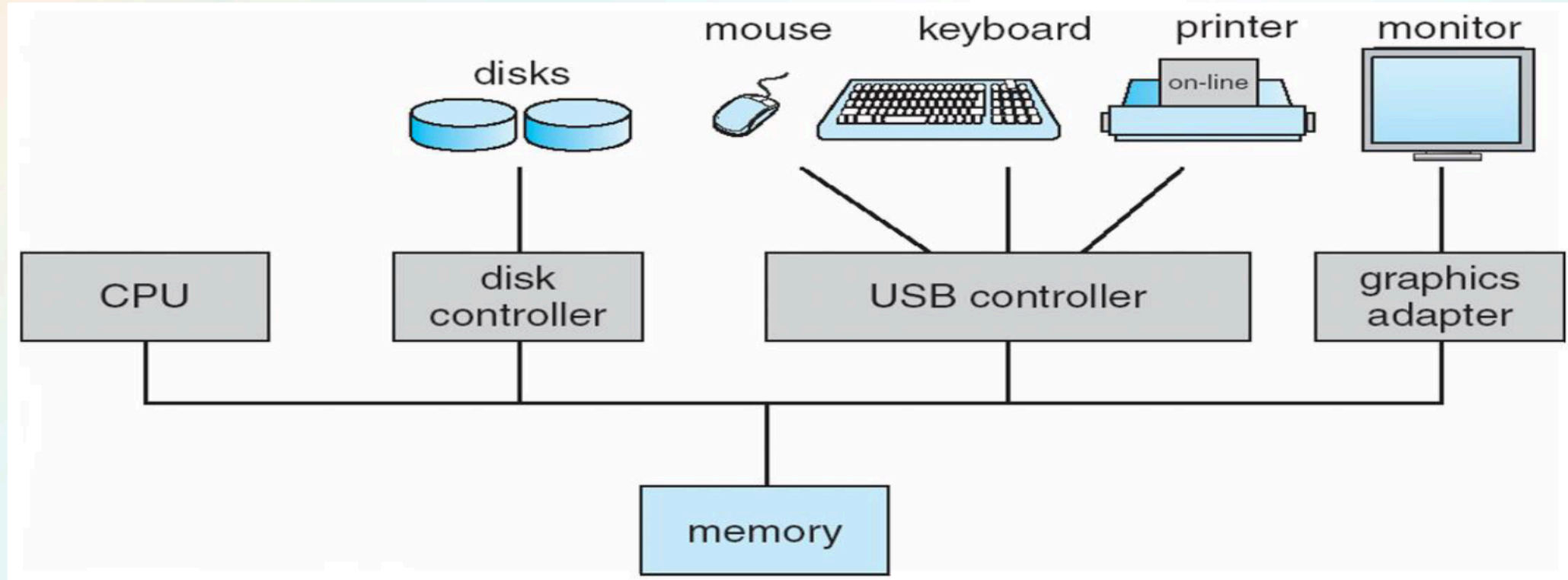
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COMPUTER SYSTEM ORGANIZATION



COMPUTER SYSTEM OPERATION

- One or more CPUs, device controllers connect through common **bus** providing access to shared memory.
- Concurrent execution of CPUs and devices competing for memory cycles.
- I/O devices and the CPU can execute concurrently.
- Each device controller is in charge of a particular device type.
- Each device controller has a local buffer.
- CPU moves data from/to main memory to/from local buffers.
- I/O is from the device to local buffer of controller.
- Device controller informs CPU that it has finished its operation by causing an ***interrupt***.



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WHAT IS AN INTERRUPT?

- **Interrupt** is a signal emitted by hardware or software when a process or an event needs immediate attention.
- It alerts the processor temporarily to a **high priority process** requiring interruption of the current working process and then return to its previous task.
- **Types of Interrupts:**
 - Hardware interrupt
 - Software interrupt
- An operating system is **interrupt driven**.



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TYPES OF INTERRUPTS

- **Hardware interrupt** is a signal created and sent to the CPU that is caused by some action taken by a hardware device.

Example: When a key is pressed or when the mouse is moved.

- **Software Interrupt** arises due to illegal and erroneous use of an instruction or data. It often occurs when an application software terminates or when it requests the operating system for some service.

Example: stack overflow, division by zero, invalid opcode, etc.
These are also called **traps**.

INTERRUPT HANDLING

- The operating system preserves the state of the CPU by storing registers and the program counter
- Determines which type of interrupt has occurred:
 - **Polling** – operating system sends signal to each devices asking if they have a request
 - **Vectored** interrupt system – requesting device sends interrupt to the operating system.
- Separate segments of code determine what action should be taken for each type of interrupt



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OPERATING SYSTEM OPERATIONS

- **Dual-mode** operation allows OS to protect itself and other system components
 - **User mode** and **kernel mode**
 - **Mode bit** provided by hardware
 - Provides ability to distinguish when system is running user code(1) or kernel code(0)
 - Some instructions designated as **privileged**, only executable in kernel mode
 - System call changes mode to kernel, return from call resets it to user
 - A **system call** is a way for programs to interact with the OS. A computer program makes a system call when it makes a request to the OS's kernel.



PROCESSOR SYSTEM

Single-Processor System

- There is one main CPU capable of executing a general-purpose instruction set, including instructions from user processes.

Multiprocessor System

- Also known as **parallel-system** or **multicore**.
- First appeared in servers and now in smartphones and tablet computers.



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COMPUTER SYSTEM ARCHITECTURE

- Most systems use a **single general-purpose processor** (PDAs through mainframes)
- Most systems have special-purpose processors as well
- Multiprocessors systems growing in use and importance
 - Also known as **parallel systems** or **tightly-coupled systems**
 - Advantages include
 - Increased throughput
 - Economy of scale
 - Increased reliability – graceful degradation or fault tolerance
 - **Two types**
 - Asymmetric Multiprocessing
 - Symmetric Multiprocessing



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MULTIPROCESSOR SYSTEMS

Advantages:

1. **Increased throughput.** Increasing the number of processor, expect to get more work done in less time.
2. **Economy of scale.** It can cost less than equivalent multiple single-processor systems because they can share peripherals, mass storage and power supplies.
3. **Increased reliability.** Functions can be distributed properly among several processors. If one processor fails, the other processor can pick-up the task.



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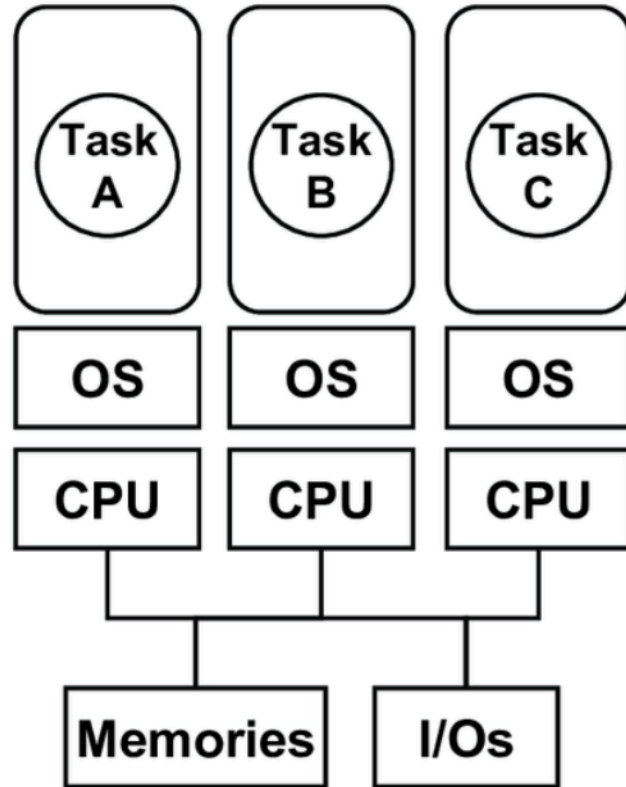
MULTIPROCESSOR SYSTEMS

The multiprocessor systems in use today are of two types.

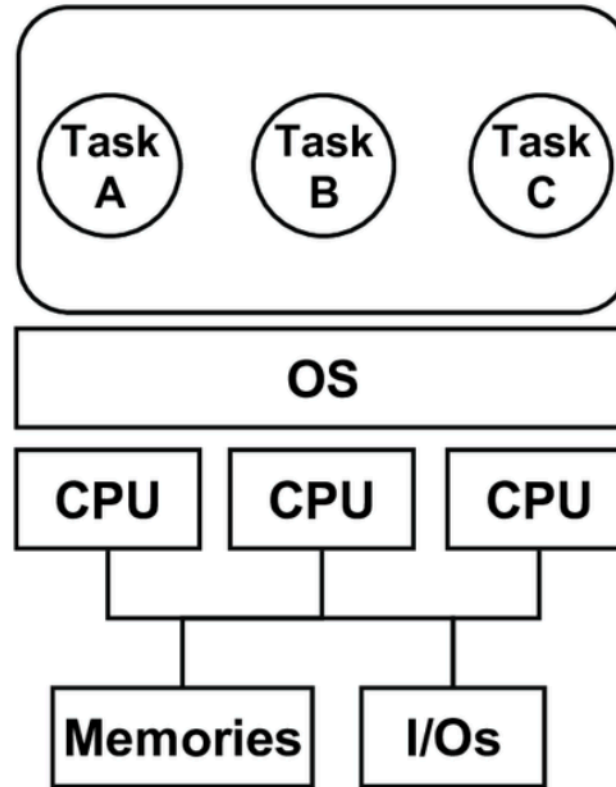
1. **Asymmetric multiprocessing.** Each processor is assigned a specific task. A boss processor controls the system and the other processors either look to the boss for instructions or have predefined tasks. **Boss-worker relationship.**
2. **Symmetric multiprocessing(SMP).** The most commonly used. In which each processor performs all tasks within the operating system. **All processors are peers and no boss-worker relationship.**

The difference between symmetric and asymmetric may result from either hardware or software.

MULTIPROCESSOR SYSTEMS



(a) Asymmetric
Multi-Processing (AMP)

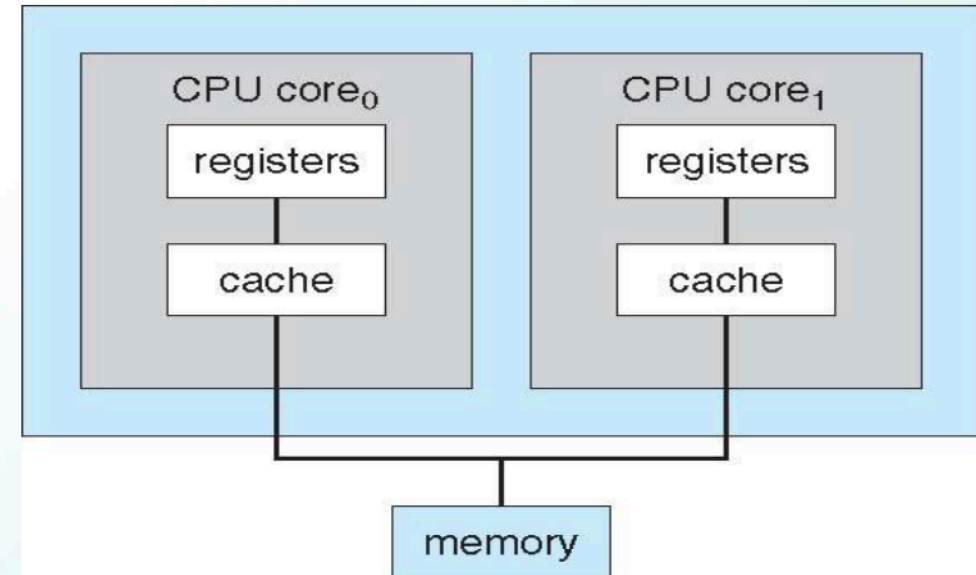


(b) Symmetric
Multi-Processing (SMP)



MULTIPROCESSOR SYSTEMS

- A recent trend in CPU design is to include multiple computing cores on a single chip. Such multiprocessor systems are termed **multicore**. They can be more efficient than multiple chips with single core.
- A dual-core design with two cores on the same chip. Each core has its own register set as well as its own local cache.



DUAL CORE DESIGN

CLUSTERED SYSTEMS

- Like multiprocessor systems, but multiple systems working together
- Usually sharing storage via a **storage-area network (SAN)**
- Provides a high-availability service which survives failures
 - **Asymmetric clustering** has one machine in hot-standby mode
 - **Symmetric clustering** has multiple nodes running applications, monitoring each other
- Some clusters are **for high-performance computing (HPC)**



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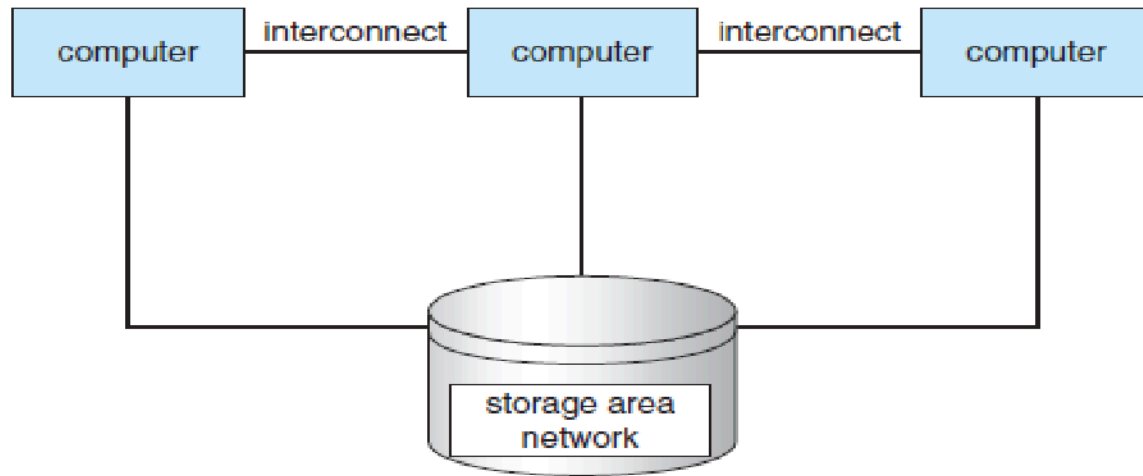
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CLUSTERED SYSTEM STRUCTURE



NAS vs. SAN Storage

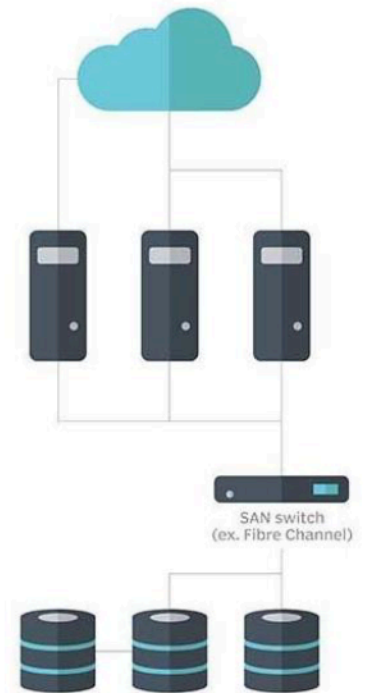
NETWORK-ATTACHED STORAGE

- Shared storage over shared network
- File system
- Easier management



STORAGE AREA NETWORK

- Shared storage over dedicated network
- Block storage
- Fast, but expensive



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COMPUTING ENVIRONMENT

- **Traditional computer** - blurring over time
 - Office environment
 - PCs connected to a network, terminals attached to mainframe or minicomputers providing batch and timesharing
 - Now portals allowing networked and remote systems access to same resources
 - Home networks
 - Used to be single system, then modems
 - Now firewalled, networked



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COMPUTING ENVIRONMENT

- **Mobile computing**
 - Refers to computing on handheld smartphones and tablet computers.
- **Distributed system**
 - It is a collection of physically separate, possibly heterogeneous computer systems that are networked to provide users with access to the various resources that the system maintains.
 - Network operating system is an operating system that provides services across the network.



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COMPUTING ENVIRONMENT

- **Client-Server Computing**
 - Dumb terminals succeeded by smart PCs
 - Many systems now **servers**, responding to requests generated by **clients**
 - **Compute-server** provides an interface to client to request services (i.e. database)
 - **File-server** provides interface for clients to store and retrieve files



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COMPUTING ENVIRONMENT

- Another model of distributed system
- **P2P** does not distinguish clients and servers
 - Instead **all nodes are considered peers**
 - May each act as client, server or both
 - Node must join P2P network
 - Registers its service with central lookup service on network, or
 - Broadcast request for service and respond to requests for service via **discovery protocol**
- **Examples:** *Napster* and *BitTorrent*



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COMPUTING ENVIRONMENT

- **Virtualization**
 - It is a technology that allows operating systems to run as applications within other operating system.
 - It is one member of the class software that also includes emulation. Emulation is used when the source CPU type is different from the target CPU type.
 - **Example:** virtual machine, OracleVirtualBox



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COMPUTING ENVIRONMENT

- **Cloud Computing**

- It is a type of computing that delivers computing, storage and even applications as a service across a network.
- It is a logical extension of virtualization.
 - **Public cloud** – cloud available via the Internet
 - **Private cloud** – cloud run by a company for that company's own use
 - **Hybrid cloud** – cloud that includes both public and private



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COMPUTING ENVIRONMENT

- **Cloud Computing Service Models**
 - **Software as a service(SaaS)** – one or more applications available via the Internet
 - **Platform as a service(PaaS)** – software stack ready for application use via the Internet
 - **Infrastructure as a service(IaaS)** – servers or storage available over the Internet

Platform Type	Common Examples
SaaS	Google Apps, Dropbox, Salesforce, Cisco WebEx, Concur, GoToMeeting
PaaS	AWS Elastic Beanstalk, Windows Azure, Heroku, Force.com, Google App Engine, Apache Stratos, OpenShift
IaaS	DigitalOcean, Linode, Rackspace, Amazon Web Services (AWS), Cisco Metapod, Microsoft Azure, Google Compute Engine (GCE)

OPEN-SOURCE OPERATING SYSTEM

- **Open Source operating systems** are released under a license where the copyright holder allows others to study, change as well as distribute the software to other people.
- Counter to the copy protection and **Digital Rights Management (DRM)** movement
- Started by **Free Software Foundation (FSF)**, which has “copyleft” GNU Public License (GPL)
- **Examples:** GNU(**GNU’s** Not **Unix**)/Linux, BSD UNIX (including core of Mac OS X), and Sun Solaris



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